ECG COMPASS

ECG pocket guide

11 cards for the ward, the ER and the night shift

Act now: call for help

- ST elevation, or a STEMI equivalent, with chest pain
- VT, VF, polymorphic VT or torsades de pointes
- Complete AV block, Mobitz II, or pauses over 3 s with symptoms
- Fast, irregular, wide QRS: think pre-excited AF
- Signs of hyperkalaemia
- QTc 500 ms or more
- Brugada type 1 pattern with syncope
- Any brady- or tachycardia with low blood pressure, chest pain or confusion

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Education only. Follow your guidelines and local protocols. Treat the patient, not the ECG.

1 Read every ECG the same way

1. **Check the basics.** Right patient, date and time. Calibration 10 mm/mV, speed 25 mm/s.
2. **Rate.** See card 2.
3. **Rhythm.** Regular? A P wave before every QRS? A QRS after every P?
4. **Axis.** See card 3.
5. **P wave.** Upright in I and II, negative in aVR? Size and shape.
6. **PR interval.** Short, normal or long? Constant?
7. **QRS.** Width, shape (bundle branch block?), Q waves, voltage.
8. **ST segment.** Elevated or depressed? Which leads?
9. **T wave** (and U wave). Inverted, peaked, flat?
10. **QT and QTc.**
11. **Compare** with the previous ECG and with the patient in front of you.

> **Lead check:** negative P wave and QRS in lead I with a positive aVR? Think limb electrodes swapped (or dextrocardia). Repeat the ECG.

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2 Paper, rate and normal values

1 small square	1 mm = 40 ms = 0.1 mV
1 large square	5 mm = 200 ms = 0.5 mV
5 large squares	1 second

Rate

- **Regular:** 300 ÷ large squares between two R waves (or 1500 ÷ small squares).
- Large squares 1-2-3-4-5-6 = **300-150-100-75-60-50**/min.
- **Irregular:** number of QRS complexes in the 10 s strip × 6.

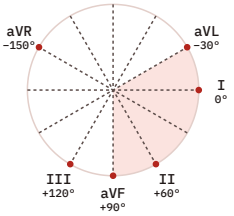
Normal adult values

Sinus rate	60–100/min
P wave	< 120 ms, < 2.5 mm in II
PR	120–200 ms
QRS	< 120 ms (usually ≤ 100 ms)
QTc	≤ 450 ms men, ≤ 460 ms women
Axis	−30° to +90°

Bazett: $QTc = QT \div \sqrt{RR}$ (RR in seconds)
Fridericia: $QTc = QT \div \sqrt[3]{RR}$ better below 60 or above 90/min

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3 Axis in 30 seconds



I	aVF	Axis
+	+	Normal
+	−	Look at II: + normal (0 to −30°), − left deviation
−	+	Right deviation
−	−	Extreme axis

Fine-tune: the lead where the QRS is most equiphasic is at 90° to the axis.

Left axis (beyond −30°)

Left anterior fascicular block, inferior MI, LBBB, LVH, right ventricular pacing, WPW.

Right axis (beyond +90°)

RVH, acute right heart strain (pulmonary embolism), left posterior fascicular block, lateral MI, dextrocardia, swapped limb electrodes. Can be normal in young or slim people.

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4 AV blocks

First degree
PR > 200 ms. Every P wave is conducted.

Second degree, Mobitz I (Wenckebach)
PR lengthens beat by beat until one P wave is not conducted. The PR after the pause is the shortest. Usually in the AV node.

Second degree, Mobitz II
Constant PR, then a P wave suddenly not conducted. Usually below the AV node, often with a wide QRS. Usually needs a pacemaker.

2:1 block
Every second P wave not conducted. One strip can't tell type I from type II; a wide QRS points below the node.

Third degree (complete)
P waves and QRS complexes beat independently; more P waves than QRS. Escape rhythm narrow (junctional, 40–60/min) or wide (ventricular, 20–40/min).

Look for hidden P waves in the T waves and the QRS. March them out with calipers.

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5 Bundle branch and fascicular blocks

Right bundle branch block QRS ≥ 120 ms

- rsR′ ("M") in V1–V2
- Wide, slurred S wave in I, aVL, V5–V6
- QRS 110–119 ms with this pattern: incomplete RBBB

Left bundle branch block QRS ≥ 120 ms

- QS or rS in V1
- Broad, notched or slurred R in I, aVL, V5–V6
- No septal q waves in I, V5–V6
- ST–T goes opposite to the QRS (this is expected)

Left anterior fascicular block QRS < 120 ms

Axis −45° to −90°. qR in aVL, rS in II, III and aVF.

Left posterior fascicular block QRS < 120 ms

Axis +90° to +180°. rS in I and aVL, qR in III and aVF. Rule out other causes of right axis first.

Bifascicular block: RBBB plus either fascicular block. Add a long PR and syncope: think of conduction disease that may need pacing.

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6 STEMI: criteria and territories

New ST elevation at the J point in **2 or more contiguous leads**:

V2–V3	≥ 2.5 mm men under 40 ≥ 2 mm men 40 and over ≥ 1.5 mm women
Other leads	≥ 1 mm
V7–V9	≥ 0.5 mm (posterior)
V3R–V4R	≥ 0.5 mm (≥ 1 mm men under 30)

ST ELEVATION IN	WALL	USUAL ARTERY
II, III, aVF	Inferior	RCA, sometimes LCx
V1–V4	Septal, anterior	LAD
I, aVL, V5–V6	Lateral	LCx or diagonal
V4R, with inferior MI	Right ventricle	Proximal RCA
ST depression V1–V3, tall R in V1–V2	Posterior	LCx or RCA: record V7–V9

Reciprocal ST depression in the opposite leads supports a true STEMI.

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7 Don't miss: equivalents and look-alikes

Treat like STEMI

Posterior MI

ST depression in V1–V3 with upright T waves. Record V7–V9.

de Winter pattern

Upsloping ST depression with tall, symmetric T waves in V1–V6. Proximal LAD occlusion.

LBBB or paced rhythm with ischaemic symptoms

Sgarbossa: concordant ST elevation ≥ 1 mm, concordant ST depression ≥ 1 mm in V1–V3, or discordant ST elevation ≥ 25% of the S wave depth (Smith).

High risk, not STEMI

Widespread ST depression + ST elevation in aVR

Left main or multivessel disease, or global subendocardial ischaemia.

Wellens pattern

Biphasic or deeply inverted T waves in V2–V3 in a pain-free patient: critical proximal LAD stenosis.

Look-alikes

Early repolarisation, pericarditis, LVH, LBBB, ventricular pacing, Brugada pattern, hyperkalaemia, LV aneurysm, Takotsubo.

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8 Tachycardias

	Regular	Irregular
Narrow QRS < 120 ms	Sinus tachycardia AVNRT AVRT Atrial flutter 2:1 (~150/min) Atrial tachycardia	Atrial fibrillation Flutter with variable block Multifocal atrial tachycardia (≥ 3 P shapes)
Wide QRS ≥ 120 ms	**VT until proven otherwise** SVT with bundle branch block Antidromic AVRT	AF with bundle branch block **Pre-excited AF** (WPW) Polymorphic VT, torsades

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Clues for VT

- AV dissociation; capture or fusion beats
- Concordance: all precordial QRS positive or all negative
- QRS > 160 ms; extreme axis (−90° to ±180°)
- Initial R wave in aVR
- No RS complex in any precordial lead

Rate around 150/min and regular: look hard for flutter waves in II, III, aVF and V1. Irregular, very fast and wide: pre-excited AF, where AV nodal blockers are dangerous.

9 Electrolytes, drugs, temperature

High potassium

Tall peaked T waves → longer PR, flat P → wide QRS → sine wave, bradycardia, VF or asystole. The ECG can look normal despite a high level.

Low potassium

Flat T waves, prominent U waves, ST depression, long QU interval.

Calcium

Low: long QT through a long ST segment. High: short QT.

Digoxin

Effect: scooped ("reverse tick") ST depression. Toxicity: almost any arrhythmia, classically atrial tachycardia with block or bidirectional VT.

Hypothermia

Osborn (J) waves, bradycardia, long intervals, shivering artefact.

Long QT

QTc ≥ 500 ms: high risk of torsades. Check drugs, potassium, magnesium and calcium.

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10 Patterns to recognise

WPW pattern

PR < 120 ms, delta wave (slurred QRS start), wide QRS, secondary ST–T changes.

Brugada type 1

Coved ST elevation ≥ 2 mm in V1–V2 (also in high positions), followed by a negative T wave.

Pericarditis

Widespread concave ST elevation, PR depression (PR elevation in aVR), no reciprocal ST depression except in aVR and V1.

Pulmonary embolism

Most often just sinus tachycardia. Also: RBBB, right axis, T inversion in V1–V4 and III, S1Q3T3 (neither sensitive nor specific).

Left ventricular hypertrophy

Sokolow–Lyon: S in V1 + R in V5 or V6 ≥ 35 mm. Cornell: R in aVL + S in V3 > 28 mm (men), > 20 mm (women). Strain: ST depression and T inversion in lateral leads.

Right ventricular hypertrophy

R ≥ 7 mm or R/S > 1 in V1, right axis, strain in V1–V3.

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11 Pacemaker ECG

- **Spikes:** before the P wave (atrial pacing), before the QRS (ventricular pacing). Bipolar leads give tiny spikes: look in every lead.
- **Right ventricular pacing:** wide QRS with an LBBB-like shape, ST–T opposite to the QRS (expected).
- **Biventricular pacing:** often an R wave in V1 and a negative QRS in I.
- **Conduction system pacing** (His or left bundle area): a narrower QRS.

Mode letters — 1: chamber paced · 2: chamber sensed (A, V, D, O) · 3: response (I inhibit, T trigger, D both, O none) · R: rate response. Examples: VVI, AAI, DDD.

PROBLEM	WHAT YOU SEE
Failure to capture	Spike with no P or QRS after it
Undersensing	Spikes where they don't belong, e.g. in the T wave
Oversensing	Pauses without spikes
No output	No spike when one is due; looks like oversensing

Magnet: on a pacemaker, fixed-rate pacing (VOO or DOO). On an ICD, it usually pauses shocks and leaves pacing unchanged. Details vary by manufacturer. Fusion beats are normal.

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